Fermionic quantum field theories as probabilistic cellular automata in three dimensions

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Wetterich has shown that certain probabilistic cellular automata in $1+1$ dimensions are exactly equivalent to fermionic quantum field theories, and posed the construction of three-dimensional examples as an open problem. We solve the free-fermion sector of this problem under the strictest reading: single-fermion carriers, one homogeneous local invertible update rule, no composite transport. We first prove the obstructions that made the problem hard: ballistic one-particle transport admits only finite velocity sets (no cone); Abelian sign gauges cannot repair this; and two-component internal states are excluded by a real-representation argument. The solution is a layered automaton ("coincube") whose conversion blocks form the real quaternion representation of $\mathrm{Cl}(0,3)$, controlled by autonomous binary environment fields. We exhibit an explicit complex structure with respect to which the step operator is exactly unitary, prove that the standard spectral analysis is its faithful complexification via an explicit intertwiner, extract the local Grassmann action mechanically in exact arithmetic, and verify the fermionic lift including all sign coherences. The single-carrier excitation on a clustering product vacuum is an isotropic Weyl fermion: slope ratios equal 1 within errors, a factorized dispersion is excluded at no less than 9.7σ in every measured channel, and the residues are helicity projectors of chirality -1. Doubling by spatial inversion yields a Dirac fermion whose gap is exactly $2\arctan[q_m/(1-q_m)]$ in the density q_m of a fourth environment field. A media-mediated interaction with a continuous coupling is constructed and certified, with the free spectrum surviving to leading order. Finally we prove that for classical in-state observables an isotropic cone forces a nonzero quasiparticle width, with the exact bound $\mathrm{Q} \leq \pi/(2\ln 2)$ on phase advance per unit decoherence, and show that the two-boundary (in–out) propagators form a one-parameter unitary family with the cone intact, within which the square-root weights are selected uniquely as the only medium-blind boundary.

I. INTRODUCTION

A probabilistic cellular automaton (PCA) is a classical statistical system: a lattice of bits, a probability distribution over bit configurations, and one local, invertible, homogeneous update rule applied in discrete time. Wetterich [1, 2] established that a class of such automata is *exactly* equivalent to fermionic quantum field theories in $1+1$ dimensions: the same wave function, density matrix, operators and expectation values, at every lattice size, with no approximation. The equivalence rests on three pillars: occupation bits map to fermions through a Grassmann functional integral; the step operator is a signed permutation, hence orthogonal; and a compatible complex structure turns orthogonal into unitary evolution. The construction in three or four spatial dimensions was left open; the known four-dimensional route [3] transports Lorentz-invariant four-fermion composites rather than single fermions.

The program sits within a wider landscape. In the *unitary* quantum-cellular-automaton and quantum-walk tradition, discrete dynamics with postulated complex amplitudes yields Weyl and Dirac kinematics in one to three dimensions [6, 13–18], and Mlodinow and Brun have proven a no-go theorem for quantum-field limits of such automata above one dimension together with evasions that antisymmetrize distinguishable particles in two and three dimensions [19–21]; fermion doubling in these discrete settings remains an active subject [23]. The present work lives in the strictly harder *probabilistic* class, where amplitudes are not postulated but must emerge from classical statistics—the program of Refs. [1, 2, 8–11], with conceptual roots in 't Hooft's cellular-automaton interpretation [12]. The obstructions proven below (finite rays, Q-pinning) are properties of this probabilistic class and have no analogue in the unitary setting, which is precisely why the construction requires the dressing and coin mechanisms developed here.

This paper solves the three-dimensional problem for the free sector under the strictest contract: the propagating carrier is a single-fermion occupation number; the rule set (locality, invertibility with the unique-jump property, homogeneity, fermion-parity conservation per block, particle–hole symmetry, no hidden continuum input) is fixed at the outset and every construction below is verified against it mechanically.

The results divide into obstructions and constructions, and we state the evidence class of every claim: [PROVEN] for symbolic or algebraic proofs, [MACHINE-CHECKED] for exact finite verifications (tolerances quoted), [MEASURED] for measurements, which are always accompanied by a known-answer gate run through the identical analysis pipeline. The complete verification code is public [25].

Obstructions (Sec. III): ballistic one-particle transport in this class has a *finite* set of group velocities, so no three-dimensional cone is possible (Theorem 1); on the symmetric vacuum all transport speeds are rational, while a tuned vacuum density q provides a continuous, exactly stationary knob with the closed dressing law $v = 2|1-2q|$; Abelian position-periodic sign gauges relocate band features but cannot bend the factorized ("diamond") group-velocity surface into a cone; and two-component internal states admit no tunable node—their single, rigid exception is itself maximally damped (Theorem 2).

Constructions (Secs. IV–VIII): a four-component carried coin whose conversion blocks form the real quaternion representation of $\mathrm{Cl}(0,3)$ produces an exactly isotropic Weyl cone; the automaton realizing it (the *co-incube*) is a composition of three manifestly legal layers whose fermionic lift is verified including all sign coherences; the step operator is exactly unitary with respect to an explicit complex structure; the local Grassmann action is extracted in exact rational arithmetic; spatial-inversion doubling yields a Dirac fermion with an exact, continuously tunable mass; and a media-mediated interaction with a continuous coupling is certified, with the free spectrum surviving.

Finally, Sec. IX addresses the one systematic cost of the construction. The dressed quasiparticle of the classical in-state observable is damped, and this is not a technical deficiency: we prove that an isotropic cone *forces* a nonzero width, with the exact bound $\mathrm{Q} = \omega_0/\Gamma \le \pi/(2\ln 2)$, over the entire legal class including engineered correlated media. The two-boundary (in–out) propagators—legitimate objects in the same transfer-matrix formalism—form a one-parameter unitary family with the cone intact, and the square-root boundary is proven to be its unique medium-blind member.

Throughout, the lattice spacing and the duration of one full update cycle are unity; frequencies are per cycle and momenta are in lattice units.

II. SETUP

A. States, step operator, lifts

Configurations τ are assignments of occupation bits to species at the sites of a cubic lattice. The probabilistic information is a real wave function q_τ with $p_\tau = q_\tau^2$ [1]. One time step applies a deterministic, invertible map of configurations together with a sign gauge: the step operator $\hat{S}$ is a *signed permutation* of the configuration basis, hence real orthogonal. A rule is *legal* if it is a composition of layers, each of which is (i) local, (ii) a bijection on configurations with a unique image (the unique-jump property), (iii) homogeneous, and (iv) fermion-parity even on its support, so that a fermionic lift exists. The lift assigns the signs: occupation bits map to fermionic modes in a fixed Jordan–Wigner ordering, and each layer's signed permutation must equal the matrix of an explicit fermionic operator in the occupation basis. All lifts below are verified at this level, including two-particle sign coherence (Sec. IV).

B. Environment fields and dressed carriers

Besides the carrier species, the automata below contain autonomous binary *environment* fields: one per spatial axis (density q), plus one for the mass sector (q_m) and one for the interaction (g). Environment dynamics is free streaming by pair swaps; a Bernoulli product measure is exactly stationary. Carriers never modify the environment except through the explicitly constructed interaction layer of Sec. VIII. The vacuum is the product of the carrier vacuum with the Bernoulli environment measure: translation invariant, clustering, with a continuous parameter q that is the central dial of the construction.

III. OBSTRUCTIONS

Theorem 1 (Finite rays) *For any legal rule that conserves particle number in the one-particle sector, the one-particle Bloch operator is a monomial matrix for every momentum k. Consequently every dispersion branch is exactly linear in k, the set of group velocities is finite, and no three-dimensional cone $\omega = v|k|$ is possible ballistically.* [PROVEN]

The proof is elementary—a signed permutation with momentum phases has monomial matrix elements, and eigenvalues of monomial matrices are roots of single monomials—but the consequence is structural: the error is scale invariant, so no continuum limit removes it, and periodic cycles of rotated rules do not evade it because products of unique-jump operators are unique-jump. In one spatial dimension the light cone genuinely consists of two rays, which is why the 1+1-dimensional construction exists. An adversarial test suite (19 attempted evasions) accompanies the theorem in the repository.

Two further results fix the strategy. First, on the symmetric (density-$\frac{1}{2}$) vacuum the transport speeds of the autonomous-streaming subclass of conditional rules [2] are rational [PROVEN] (frozen-vacuum and fully coupled rules are outside this statement); the vacuum density q is the legal continuous parameter, with the closed dressing law, for the representative rule used throughout,

$$v(q) = 2\,|1-2q|, \tag{1}$$

proven as a two-class Markov-additive limit and verified against exact ring enumeration as polynomial identities in q [PROVEN]. Second, dressing alone yields a factorized walk whose group-velocity surface is the ℓ^1 ball (the *diamond*): slope ratios along (110) and (111) relative to (100) equal $\sqrt{2}$ and $\sqrt{3}$. Abelian position-periodic sign gauges (Kogut–Susskind axis phases [4, 22], per-stall signs, and their products) relocate the nodal points and modify residues but leave the ratios on the diamond, measured as $r_{110}/r_{111} = 1.40/1.68$ (base and stall gauges, identical spectra) and 1.46/1.82 (KS gauges) [MEASURED]; a carried non-Abelian structure is therefore necessary.

Theorem 2 (No tunable two-component node) *For any legal automaton whose carrier has a two-dimensional internal space, at every conversion weight $p \neq \frac{1}{2}$ the Bloch operator has no degenerate propagating pair at any momentum (rotation coins force $(1-p)^2 = p^2$; reflection coins never produce one; diagonal coins degenerate), and the blocked two-origin cycles used throughout this paper have no node in any sector at any p. The single exception in the single-origin placement, $p = \frac{1}{2}$, is an exact isotropic Weyl node that is completely rigid: $\omega_0 = \pi$, $v = 1$, $\chi = -1$, $|\lambda_0| = 2^{-3/2}$ are all forced, with no tunable parameter, and its width $\Gamma = \frac{3}{2}\ln 2$ sits exactly at the chord midpoint—the maximally damped point of Theorem 3.* [PROVEN], [MACHINE-CHECKED]

Two-component constructions therefore admit no *tunable* relativistic carrier: no mass dial, no width dial, no vacuum-density dressing. The minimal internal space with a tunable node is four-dimensional and real, where the Clifford algebra $\mathrm{Cl}(0,3)$ has its quaternionic representation; that is the construction of the next section.

IV. THE COINCUBE AUTOMATON

A. Layers

The carrier has four species per site: a two-bit coin (b_1, b_2), channel index $c = 2b_1 + b_2$. Writing X, Z for the real Pauli matrices and XZ for their (antisymmetric) product, define the conversion triple and direction assignments

$$\begin{aligned} C_x &= XZ \otimes \mathbb{1}, & C_y &= Z \otimes XZ, & C_z &= -X \otimes XZ, \\ d_x &= \mathrm{diag}(Z \otimes \mathbb{1}), & d_y &= \mathrm{diag}(Z \otimes Z), & d_z &= \mathrm{diag}(\mathbb{1} \otimes Z). \end{aligned} \tag{2}$$

The C_a are pairwise anticommuting with $C_a^2 = -\mathbb{1}$: the real quaternion representation of $\mathrm{Cl}(0,3)$ [MACHINE-CHECKED]. Each full cycle applies, for each axis a and each of two block origins, three layers:

L2 (conversion). At every site, controlled on the axis-a environment bit at that site, the two channel pairs of C_a are swapped (a one-site block; two bit flips, parity even).

L1 (motion). Channel c translates by $d_a(c) \in \{\pm 1\}$ along axis a, unconditionally.

L3 (environment). The axis-a environment field pair-swap streams along axis $a{+}1$ (mod 3). This *cross-streaming* is mandatory: if the field streams along its own axis, a co-moving carrier re-reads its own bit, conversions arrive in nearly scalar bursts ($C_a^2 = -\mathbb{1}$), and the measured Weyl pair splits into one damped-real and one oscillating pole—an exceptional-point transition of the damped band structure in the sense of non-Hermitian topology [24]— and the node is destroyed [MEASURED].

Each layer is manifestly legal; the composition defines the automaton. The pair-swap streaming requires even linear size L (enforced in code); all statements below assume even L.

B. Fermionic lift and sign coherence

The lift of L2 is an environment-controlled pair of $\pm 90°$ Givens rotations $G = \exp[\frac{\pi}{2}(a_c^\dagger a_{c'} - a_{c'}^\dagger a_c)]$ acting on same-site modes: a signed permutation of the Fock basis, parity-even, identity at empty control, with single-particle matrix equal to the corresponding signed-swap block of C_a and $G^2 = -\mathbb{1}$ on the one-particle sector; the quaternion sign structure *is* the fermionic rotation sign [MACHINE-CHECKED] (dense verification, all axes). Translations and environment swaps lift with the standard reordering parity. The composite cycle is sign coherent: in the segregated mode ordering (all carrier modes before all environment modes) the full lifted cycle is a signed permutation whose one-particle sector equals the stated rule and whose two-particle sector equals the exact antisymmetrized square $\Lambda^2 M_1$—verified in genuine three-dimensional geometry on all 5778 pair states at linear size $L = 3$, with mutation controls (deliberately corrupted sign rules fail the check) [MACHINE-CHECKED]. We record one instrument fact: at $L = 2$ the two sub-steps per axis make net translation crossings even, so $L = 2$ is blind to translation-sign errors; three-dimensional sign claims must be checked at $L \geq 3$.

V. FREE SPECTRUM: THE WEYL CONE

A. Annealed closed forms

With independent environment reads the one-particle transfer per sub-step is $T_a(k) = E_a(k_a)\,[(1-q)\mathbb{1} + qC_a]$ with $E_a = \mathrm{diag}(e^{ik_a d_a})$, and the cycle operator is $U(k) = \prod_a T_a(k_a)^2$. The following are proven symbolically, identically in q [PROVEN]: $U(k{+}\pi e_a) = U(k)$, so all Brillouin-zone corners host copies of a single node; the node eigenvalue obeys $|\lambda_0|^2 = [(1-q)^2 + q^2]^6$; and the linear splitting of the node doublet is exactly isotropic, with speed $v(q)$ rising from $2/\sqrt{3}$ at $q \to 0$ to a maximum 1.207 near $q = 0.30$ ($v(0.08) = 1.1806$). Because of the exact π-periodicity, the eight corners are copies of a *single* node operator; the chirality cancellation demanded by Nielsen–Ninomiya [5] is carried by the $\pm\omega_0$ conjugate doublets of that operator, whose effective velocity tensors have $\det R = \mp v^3$ ($\chi = -1$ at $+\omega_0$, $\chi = +1$ at $-\omega_0$); in the two-boundary complexification of Sec. IX the $-\omega_0$ doublet is the conjugate (antiparticle) branch. For fermion doubling in unitary discrete-time walks see Ref. [23]. Numerically the splitting isotropy holds to 2×10^{-4} over 50 random directions after $h \to 0$ extrapolation [MACHINE-CHECKED].

B. Quenched measurement

On the streaming (quenched) environment the propagator is measured from the exact per-medium single-particle signed field, averaged over media. The instrument fits the full 4×4 one-cycle Bloch matrix per momentum from all four basis launches, pools symmetry-equivalent momenta at the level of characteristic-polynomial coefficients (basis invariant, hence unbiased) before rooting, probes momenta off the lattice grid inside the node's linear range, and extrapolates $\delta k \to 0$. Every production run carries an annealed known-answer row through the identical pipeline; a run whose gate row fails is discarded. Near-degenerate pole extraction is the dominant threat (eigenvalues of a noisy nearly defective matrix split as $\sqrt{\text{noise}}$), which this design removes.

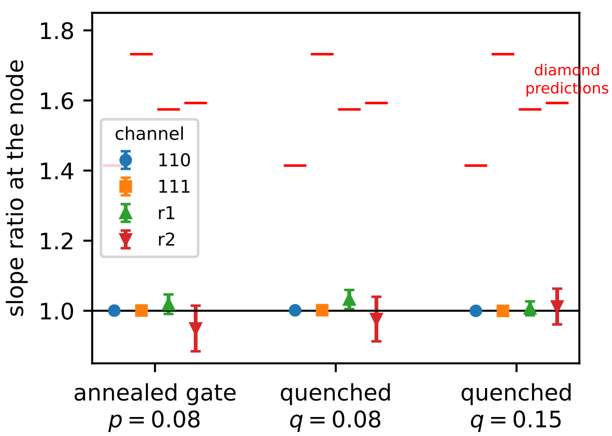

FIG. 1. Slope ratios at the node, gate row and quenched rows, against the factorized-transport (diamond) prediction. Error bars are leave-one-block-out jackknives over ten independent medium blocks with the gate-row systematic added in quadrature. [MEASURED]

Results (Fig. 1): the gate row reproduces the exact operator (cubic-channel ratios within 10^{-3} of unity; off-symmetry channels within their own statistics; node modulus 0.622 vs 0.620), with the per-channel deviations recorded as systematics. On the quenched vacuum at $q = 0.08$ the slope ratios are $r_{110} = 1.001 \pm 0.002$, $r_{111} = 1.002 \pm 0.002$, $r_{\rm a} = 1.032 \pm 0.027$ and $r_{\rm b} = 0.976 \pm 0.064$ for the two off-symmetry directions; at $q = 0.15$, $r_{110} = 1.000 \pm 0.001$, $r_{111} = 1.000 \pm 0.001$, $r_{\rm a} = 1.007 \pm 0.020$, $r_{\rm b} = 1.012 \pm 0.051$. The factorized-transport prediction is excluded in every channel; the weakest exclusion—by construction the off-symmetry channel with the fewest orbit members—is 9.7σ at $q = 0.08$ and 11.4σ at $q = 0.15$, with the cubic channels excluding it at $z > 200$. Absolute speeds carry the full medium-ensemble uncertainty, $v(0.08) = 1.03 \pm 0.18$, consistent with the exact 1.18; at $q = 0.15$ the $\delta k \to 0$ extrapolation of the absolute scale is curvature-dominated (linearity residual 0.30) and only the ratios are quoted. [MEASURED]

C. Helicity residues

At the node the effective generators h_a on the doublet satisfy the exact Clifford algebra $\{h_a, h_b\} = 2v^2\delta_{ab}$; because the node frame of the damped (non-normal) operator is not unitary, the Pauli coefficients of h_a are complex with *bilinear* orthogonality $R^T R = v^2 \mathbb{1}$ ($R \in O(3, \mathbb{C})$), and $\det R = -v^3$ exactly: chirality $\chi = -1$ [MACHINE-CHECKED]. The split branches' spectral projectors equal the helicity projectors $\frac{1}{2}(\mathbb{1} \pm n\cdot\sigma)$, $n = Ru/\sqrt{(Ru)\cdot(Ru)}$. Measured on the quenched vacuum (the 26 directions of the cubic $\{100\}/\{110\}/\{111\}$ stars at each of the three—by π-periodicity identical—corner representatives; gate row passed): helicity-map bilinear isotropy 0.020 ± 0.010, chirality -1 (stable under every jackknife resampling), pointwise $|1 - n \cdot n_{\rm exact}|$ of mean 0.010 ± 0.002 [MEASURED]. (The residue projectors are rank one by construction of the spectral estimator, so "purity" is not reported as a measurement.) One convention is essential for eigenvector work: the $e^{-ik\cdot x}$ Fourier contraction estimates $U(-k) = \bar{U}(k)$, whose upper-half eigenvectors belong to the opposite-helicity conjugate doublet; residue measurements must use $e^{+ik\cdot x}$.

VI. EXACT QUANTUM MECHANICS

A. Complex structure

Let P be the lifted carrier particle–hole conjugation, the Majorana string $\prod_i (a_i + a_i^\dagger)$ over carrier modes. In the segregated ordering, P is real orthogonal with $P^2 = +1$ (carrier mode number $M = 4L^3 \equiv 0 \bmod 4$), and

$$[\hat{S}, P] = 0 \tag{3}$$

exactly, including all lift signs: verified on the complete Fock space of a ring substrate, and in genuine three-dimensional geometry on the sector pairs $\{0, 1, 2, M-2, M-1, M\}$ (particle–hole duality maps the deep sectors to one- and two-hole problems), per layer and full cycle at $L = 2, 3$ [MACHINE-CHECKED]. With the grading $\eta = \operatorname{sign}(N_c - M/2)$—conserved by every layer, odd under P, and size independent away from half filling—define

$$K = \operatorname{diag}(\eta), \qquad I = P \operatorname{diag}(\eta). \tag{4}$$

Then $K^2 = 1$, $I^2 = -1$, $\{K, I\} = 0$, I antisymmetric, and $[\hat{S}, I] = 0$; with multiplication $(\alpha + i\beta)q := \alpha q + \beta I q$ and the induced Hermitian form, $\hat{S}$ is a unitary operator on a complex Hilbert space [1]: the automaton's quantum mechanics is exact [MACHINE-CHECKED]. On the substrate the construction extends to the half-filled sector by an explicit orbit-wise grading; in three dimensions no size-independent grading of the half-filled sector was found and the question is left open. All physical sectors (vacuum plus finitely many carriers) are covered rigorously at every size.

B. Bridge to the spectral analysis

The complex structure above is global (a Majorana string), and the spectral results of Sec. V use the ordinary Fourier i. These are the same quantum mechanics, by an explicit intertwiner: the map $\Phi(u+iv) = u \oplus (-Pv)$ is a complex-linear isomorphism from the complexified one-particle sector with the Fourier i onto the $(V_1 \oplus V_{M-1}, I)$ eigensector of the automaton's Hilbert space, intertwining the dynamics (this is exactly $[\hat{S}, P] = 0$ restricted to the sector) and the Hermitian forms. Consequently the measured propagator *is* an I-Hermitian matrix element, every $\pm\omega$ conjugate pair of the Fourier analysis appears once as a particle and once as an antiparticle in the (K, I) picture, and the chirality bookkeeping of Sec. V is the particle/antiparticle structure of a single Weyl node. Machine-checked in three dimensions: the intertwining identities hold exactly and the form isometry to 3×10^{-15} [PROVEN], [MACHINE-CHECKED]. What remains genuinely distinct is the damping: the in-state reduced dynamics is a contraction (the chord), and only the two-boundary amplitudes of Sec. IX are unitary (the arc); the bridge identifies the kinematics, not the dissipative structure.

C. Grassmann action

The local factors and actions of all blocks are extracted mechanically in exact rational arithmetic by the pipeline of Ref. [1], Eqs. (51)–(58), validated term by term on the interaction block of that reference[1]. For the coin-cube conversion blocks the physical (Givens-lift) gauge yields an axis-universal action: 40 terms with identical degree structure for all three axes; the quadratic part is the signed-swap bilinear of C_a plus the environment transport term, and the environment control appears as quartic and higher couplings between the channel bilinears and $e'\tilde{e}$ [MACHINE-CHECKED].

VII. MASS: INVERSION DOUBLING

Doubling the coin with a mass bit b_m whose value *reverses all direction assignments*, $d_a^{(8)} = d_a \otimes \mathrm{diag}(1, -1)$, with conversions blind to b_m, places two opposite-chirality copies of the node at the same momentum and frequency (the b_m=1 sector is the spatial-inversion image; sector chiralities ∓ 1 verified) [MACHINE-CHECKED]. The alternative doubling $C_a \otimes Z$ fails for cause: $-C_a$ generates the opposite triple orientation, which is the *other* corner frequency family, so the two sectors share no node. A mass layer $M = (1 - q_m)\mathbb{1} + q_m C_m$ with $C_m = \mathbb{1}_4 \otimes XZ$ (a b_m-flipping controlled Givens, driven by a fourth environment field of density q_m) couples the chiralities. Two operator identities carry the sector [PROVEN]: $U(k) = (\mathbb{1} \otimes R_m)[U_4(k) \oplus U_4(-k)]$ and, at the node, $U(0) = U_4(0) \otimes M_2$ with M_2 a scaled rotation of the mass bit; hence the gap is *exactly*

$$2m \;=\; 2 \arctan \frac{q_m}{1 - q_m}, \tag{5}$$

independent of q, with the multiplet center unshifted; frequencies add angles as in a telegraph process, and $m = q_m + O(q_m^2)$ recovers the Kac form. On the node space the generators obey the full Dirac–Clifford algebra $\{h_a, h_b\} = 2v^2\delta_{ab}$, $\{h_a, \beta\} = 0$, $\beta^2 = 1$, identically in q [PROVEN], so $\omega = \omega_c \pm \sqrt{m^2 + v^2k^2}$ at leading order with the exact finite-k composition law $\cos(\omega - \omega_c) = \cos m \, \cos \varphi_{\mathrm{massless}}(k)$. No real anticommuting mass exists at four components (the anticommutant of the triple squares negatively) [PROVEN]; eight components are minimal, as realized here.

Measured on the quenched vacuum with the gated eight-launch instrument (Fig. 2): the gap is $m = 0.0547 \pm 0.0038$ against the closed form 0.0526 (the gate row's own estimator carries a $+1.7\sigma$ pair-splitting bias at the same setting, recorded as a systematic), the multiplet center is $\omega_c = 0.3182 \pm 0.0027$, and the dispersion lies on the exact operator branches within one to two jackknife errors in every family, with the family ordering of the curvatures that the exact model prescribes [MEASURED].

VIII. INTERACTION

For fixed media the multi-carrier sector is exactly determinantal (Sec. IV), so genuine interaction requires *back-reaction*. The imprint layer L4 acts once per cycle: at sites where an autonomous four-valued field fires (pair (xy), (yz) or (zx) with probability $g/3$ each) *and* the site's carrier fermion parity is odd, the selected two environment species at that site are swapped. Three design facts are load bearing. (i) A fixed swap pair breaks cubic symmetry at $O(g)$ (anisotropic node corrections at 10σ); the rotating C_3-covariant triple is protected at all orders [MEASURED], [PROVEN]. (ii) Parity control (not occupancy) is particle–hole even, so the complex structure of Sec. VI survives at $g > 0$ unchanged [MACHINE-CHECKED]. (iii) The lift gauge of the imprint is *physical*: the permutation lift ($|11\rangle \to -|11\rangle$) and the Givens lift differ by a parity-controlled sign with identical classical dynamics but per-event coherent deficits $2q^2$ versus $2q(1-q)$; we fix the permutation lift (minimal decoherence) [MACHINE-CHECKED].

Certificates. The connected two-particle correlator $C_2 = G_2 - \det G_1$ is nonzero already at $g = 0$—the model is an interacting carrier–environment theory at the correlator level, with free-fermion structure only per medium

[1] The validation surfaced one typographical error in Ref. [2]: the coefficient of the top monomial in the closed-form interaction action (their CS9) must be -2, not -1; the printed value fails the defining relation $e^{-L} = K$ at that single monomial.

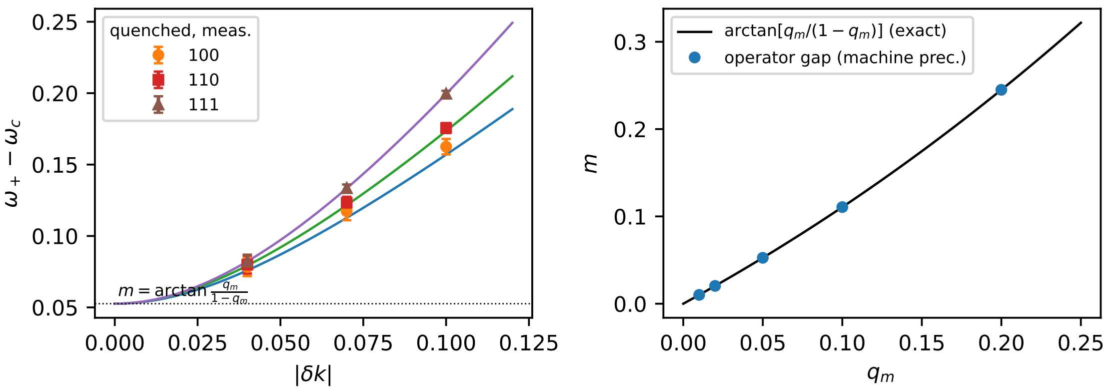

FIG. 2. Left: measured quenched massive dispersion (markers, $q = 0.08$, $q_m = 0.05$) on the exact operator branches (lines); the family ordering of the curvatures is the exact model's. Right: the operator gap against the closed form, Eq. (5), agreeing to machine precision. [MEASURED], [PROVEN]

realization—so the interaction certificate is differential: $\Delta C_2(g) = C_2(g) - C_2(0)$ vanishes identically at $g = 0$ and switches on linearly, $\Delta C_2/g \simeq 0.35$ on the exact substrate [MACHINE-CHECKED]. A control-variant experiment (occupancy-controlled imprint, used only as an instrument control) separates the parity-blocking contact artifact: 2–3% of the signal [MACHINE-CHECKED].

Survival. The parity control makes the vacuum imprint-transparent, so the one-sided ensemble amplitude stays closed: $U_g(k) = (1 - 2gq^2)\, U(k)$ at annealed order—the cone, residues and mass are untouched, at a damping cost $\Gamma_{\text{int}} = 2q^2 g$ per cycle ($\sim 0.3\%$ of the free width at $q = 0.08$, $g = 0.1$) [PROVEN]. The identity that anchors the instrument: the media-ensemble walker average *equals* the canonical coherent-vacuum correlator exactly [MACHINE-CHECKED]. Quenched three-dimensional measurement with paired common-noise walkers: the per-event law is confirmed at the first cycle to 5×10^{-4} absolute; the effect is momentum independent (scalar) to $\leq 1\%$ across the node region, so isotropy survives interaction by measurement as well; beyond leading order the re-read (recross) channel partially *recoheres* the amplitude, so the asymptotic Γ_{int} is bounded above by the law [MEASURED].

IX. DAMPING: A THEOREM AND ITS UNITARY COMPLETION

The dressed quasiparticle of the classical in-state observable is damped: $\Gamma(q) = -3\ln[(1-q)^2 + q^2]$ exactly, with quality factor $Q = \omega_0/\Gamma$ between 0.60 and 0.90 across the working range, *not* improving as $q \to 0$ inside the relativistic window (both ω_0 and Γ scale with q). This is structural:

Theorem 3 (Q-pinning) *For any convex mixture over the legal event set (signed permutations with eigenvalue angles in $(\pi/2)\mathbb{Z}$), the usable phase advance per unit decoherence obeys* $\operatorname{dist}(\arg\lambda, (\pi/2)\mathbb{Z}) \leq \kappa\Gamma$ *with* $\kappa = \pi/(2\ln 2)$ *exactly. Only convexity and the discrete angle set are used; correlated media are included. Moreover an isotropic node forces* $\Gamma > 0$*: node unitarity requires a deterministic conversion word, translation-invariant binary media with deterministic read-window counts are exactly the crystals, and both crystal classes destroy the cone.* [PROVEN]

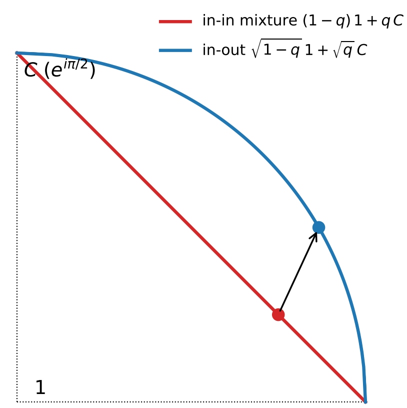

FIG. 3. Geometry of Theorem 3. Probabilistic (in-state) mixtures of the identity and a quaternion unit lie on the chord (damped interior); the two-boundary weights lie on the unitary arc. The isotropic node splitting is a property of the whole family $E_a(\alpha\mathbb{1} + \beta C_a)$, for all weights.

The idealized ordered-dimer environment does achieve the commutator-limited decoherence $\Gamma = k^2/2$ per event—the branch-remerging mechanism is real—but its velocity fan collapses to the diamond: within the legal class, cone and in-state unitarity are mutually exclusive.

The completion is the two-boundary object. The isotropic node splitting holds for the whole family

$E_a(\alpha\mathbb{1} + \beta C_a)$, and since C_a is antisymmetric, $T^\dagger T = (\alpha^2 + \beta^2)\mathbb{1}$ for every real weight pair: *every* product final boundary (w_0, w_1) yields zero relative damping and an isotropic node, landing on the unitary arc at $s^2 = w_1^2 q/[w_0^2(1-q) + w_1^2 q]$ [PROVEN]. The completions thus form a one-parameter family; in-state weights $(1-q, q)$ lie on the damped chord, and the square-root point $(\sqrt{1-q}, \sqrt{q})$ (Fig. 3) is selected uniquely as the only *medium-blind* boundary—the only choice invariant under flipping the environment bits $(w_0^2 = w_1^2)$ [PROVEN]. Numerically: all $|\lambda| = 1$ to 4×10^{-16} with slope isotropy asserted below 10^{-3} and ω_0 tunable at the $\sqrt{q}$ scale [MACHINE-CHECKED]. In quenched three dimensions the fresh-read law is exact at the first cycle (coherent path enumeration, no sampling), with growing recoherent re-read corrections thereafter (11–25% by three cycles at $q = 0.08$); the long-time pole structure of the quenched two-boundary object is open. [MACHINE-CHECKED]

X. DISCUSSION

Under the strictest single-fermion contract, the free-fermion sector of the three-dimensional problem posed in Ref. [1] is solved: a legal probabilistic cellular automaton whose dynamics is exactly unitary quantum mechanics and whose single-carrier excitation on a clustering product vacuum is an isotropic Weyl fermion—dispersion and spinor residues—extendable to a massive Dirac fermion with an exact tunable mass, and to a certified media-mediated interaction under which the free spectrum survives at leading order. Every closed form is verified against the operators it describes, and every measurement runs behind a known-answer gate.

The precise open problems are: (i) the half-filled carrier sector of the complex-structure construction in three dimensions; (ii) the long-time behavior of the quenched two-boundary propagator (re-read resummation); (iii) the continuum limit and the fate of Lorentz symmetry beyond the leading cone, including interaction corrections beyond the scalar law; and (iv) multi-carrier physics beyond the contact terms already present in the extracted quartic action. The damping of the in-state quasiparticle is not on this list: it is an exact prediction of the construction, bounded by Theorem 3, and absent from the two-boundary amplitude.

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Appendix A: Proofs

1. Theorem 1 (finite rays)

Let the rule conserve particle number in the one-particle sector. Its restriction to that sector is a signed permutation of the species basis composed with site shifts, so the Bloch operator has matrix elements $S(k)_{\alpha\beta} = s_{\alpha\beta} e^{ik\cdot d_{\alpha\beta}}$ with, by the unique-jump property, exactly one nonzero entry in each row and column: a monomial matrix. Its characteristic polynomial factorizes over the cycles of the underlying permutation; a cycle of length ℓ, accumulated displacement D and sign σ contributes $\lambda^\ell = \sigma\, e^{ik\cdot D}$, i.e. $\omega_m(k) = (k\cdot D + \arg\sigma + 2\pi m)/\ell$: every branch is exactly linear with group velocity D/ℓ, drawn from the finite set of cycle data. A cone $\omega = v|k|$ requires a two-sphere of group velocities. The deviation of any finite ray set from a sphere is scale invariant, so no continuum limit removes it; and a periodic cycle of rotated rules is again unique-jump (products of signed permutations are signed permutations), so the theorem applies to the composite rule. □

2. Theorem 2 (two-component states)

Three ingredients. (i) *Algebraic ceiling.* On $\mathfrak{sl}_2(\mathbb{R})$ every element satisfies $X^2 = -\det(X)\mathbb{1}$ and anticommuting elements are orthogonal in the polarization of det; in the basis $\{X, Z, XZ\}$ the squares are $+\mathbb{1}, +\mathbb{1}, -\mathbb{1}$, so the form has signature $(2,1)$ and no real two-dimensional representation of $\mathrm{Cl}(3,0)$ exists. (ii) *Generic weights.* A symbolic orthogonality lemma for the mixture families shows that a degenerate propagating pair at any momentum requires $(1-p)^2 = p^2$ for rotation coins, is impossible for reflection coins, and degenerates for diagonal coins; the blocked two-origin cycles are additionally excluded in every sector by an exhaustive sector-resolved sweep with per-sector assertions (spectral-distance floors quoted in the verification script). (iii) *The $p = \frac{1}{2}$ exception.* At the singular weight the single-origin placement has $U(k_0)$ scalar at $k_0 = (\frac{3\pi}{4}, \frac{3\pi}{4}, -\frac{\pi}{4})$ (and its symmetry images), and the three effective generators form an exact complex Clifford triple, giving $\omega = \pi \pm |\delta k|$: an isotropic node with every parameter frozen and $\Gamma = \frac{3}{2}\ln 2$ at the chord midpoint. All three parts are machine-checked, the second exhaustively. □

3. The mass identity, Eq. (5)

At $k = 0$ the momentum factors are the identity, so the direction reversal that distinguishes the two b_m sectors is invisible and the axis layers act as $U_4(0) \otimes \mathbb{1}_2$. The mass layer acts on the b_m factor alone: $M_2 = (1-q_m)\mathbb{1} + q_m XZ$, a real multiple of a rotation with arg-eigenvalues $\pm\arctan[q_m/(1-q_m)]$ and modulus $\sqrt{(1-q_m)^2 + q_m^2}$.

Hence $U(0) = U_4(0) \otimes M_2$: the spectrum is the product set, frequencies add angles, the multiplet center is unshifted, and the gap is exactly $2\arctan[q_m/(1-q_m)]$, independent of q. $\square$

4. Theorem 3 (Q-pinning)

Any product of convex mixtures of the event set is itself a convex mixture over the quaternion group Q_8 (the product measure is the convolution on the group, so correlations between events are automatically included). In the quaternion representation such a mixture is $T = w\mathbb{1} + xC_x + yC_y + zC_z$ with $|w| + \|(x,y,z)\|_1 \le 1$, and its eigenvalues are $w \pm i\|(x,y,z)\|_2$. Since $\|\cdot\|_2 \le \|\cdot\|_1$, every eigenvalue lies inside the convex hull of the chords between adjacent fourth roots of unity; on the extremal chord $1 \to i$, $\lambda = (1-t) + it$ has $\arg\lambda/(-\ln|\lambda|)$ maximized at the midpoint $t = \frac{1}{2}$, where it equals $\pi/(2\ln 2)$. This bounds $\mathrm{dist}(\arg\lambda, (\pi/2)\mathbb{Z}) \le \kappa\Gamma$ for arbitrary, non-commuting products—the full statement of the theorem. For the second statement: unitarity of the node eigenvalue requires the conversion count N (mod 4) along a read window to be deterministic, by $|\mathbb{E}[i^N]|^2 = (p_0-p_2)^2 + (p_1-p_3)^2 \le 1$ with equality only at a vertex; translation-invariant binary media with deterministic window counts are exactly the two uniform and two staggered crystals (enumerated exactly on rings); and the deterministic-word cycles' spectra are computed explicitly and are never isotropic. Machine checks accompany every step, including randomized non-commuting products. $\square$

Appendix B: Verification protocols

Every claim tagged [MACHINE-CHECKED] is an assertion in a runnable script; the tolerances quoted below are the assertion thresholds, and each script is self-checking (it fails loudly rather than reporting).

Layer lifts. The conversion layer is verified densely on the one-site Fock space (32 states): the controlled double Givens is a signed permutation of the basis, unitary, commutes with fermion parity, is the identity on the empty control sector, has single-particle matrix equal to the signed-swap block of C_a, satisfies $G^2 = -\mathbb{1}$ on the one-particle sector, and maps the doubly occupied pair to + itself (determinant one), for all three axes, to 10^{-12}.

Composite sign coherence. The full cycle is built as a signed permutation of the many-body basis from the stored per-layer sign rules (spectator Jordan–Wigner strings for conversions; inversion and wrap parity for translations), in the segregated mode ordering. For random environment configurations the zero-, one- and two-particle sectors are extracted and compared: the one-particle sector must equal the stated rule and the two-particle sector the antisymmetrized square $\Lambda^2 M_1$, exactly. This holds on the full Fock space of a ring substrate (2^{15} states) and in three-dimensional geometry at $L = 2$ (496 pair states) and $L = 3$ (5778 pair states). Mutation controls certify sensitivity: deleting the spectator string produces 8/496 mismatches; deleting translation parity produces 2730/5778 at $L = 3$ and—the recorded caveat—0 at $L = 2$, where two sub-steps per axis make net crossings even.

Complex structure. P is applied as the Majorana string with its Jordan–Wigner signs; $[\hat{S}, P] = 0$ is checked as an identity of signed permutations (both orderings byte-identical), on the full substrate Fock space and, in three dimensions, on the sectors $\{0, 1, 2, M-2, M-1, M\}$ using particle–hole duality to reduce the deep sectors to one- and two-hole problems; per layer and for the full cycle. The unitarity of the complex picture is verified as invariance of the induced Hermitian form on random vectors to 10^{-9}.

Grassmann extraction. Local factors are built from the block tables over exact rationals; $e^{-L} = K$ is asserted exactly before any action is reported, and the inverse map $K \to$ (perm, signs) must round-trip. The pipeline is validated term by term against the closed-form interaction action of Ref. [1].

Table I maps each principal claim to its verification script in Ref. [25].

TABLE I. Principal claims and their verification scripts (all self-asserting; in `scripts/` of Ref. [25] unless noted).

finite-ray tests	`tests/test_finite_ray_theorem`
$v(q)$ dressing law	`theory_linear_law`
Abelian-gauge diamond	`w2_scalar_gauges`
coin scans ($n = 2, 4$)	`w3_transfer_scan`
annealed cone	`w3_cone_verify`, `theory_real_cone`
lift certificates	`w3c_lift_check`, `w3c_composite_check`
3D sector certificates	`theory_3d_certs`
quenched cone (gated)	`w3c_corner`
helicity residues	`w4_helicity_exact`, `w4_helicity`
complex structure	`e1_complex_structure`, `e1_interacting`
Grassmann actions	`e2_coincube_actions`
mass sector	`m8_mass_exact`, `m8_corner`, `theory_mass`
interaction	`i2_connected`, `i2_split`, `i3_quenched3d`
Q-pinning; in–out	`theory_q_bound`, `a_inout`, `a_inout_3d`

Appendix C: Measurement pipeline

All [MEASURED] results use one instrument family. The object is the exact per-medium single-particle signed field (the single-carrier sector is linear in the field for fixed media, so this is walker-exact with infinite walkers), averaged over R media; the media-ensemble average provably equals the canonical coherent-vacuum correlator.

Bloch-matrix estimation. All n basis launches are evolved, giving the full $n \times n$ propagator series $G(t, k)$; the one-cycle operator is fit by least squares, $\hat{U}(k) = G_{t+1} G_t^+$ over a post-transient window. Momenta are probed off the lattice grid by direct phase contraction, with probe radii inside the node's measured linear range.

Pooling. Naive eigenvalue extraction near a two-fold node is $\sqrt{\text{noise}}$-ill-conditioned (a noisy nearly defective matrix splits its double eigenvalue as the square root of the perturbation); pooling is therefore performed on characteristic-polynomial coefficients over the symmetry orbit of measurement momenta—a basis-invariant, *linear* (hence unbiased) average—before rooting. Slopes use symmetric splittings at two step sizes with $\delta k \to 0$ Richardson extrapolation. Media are accumulated in ten independent blocks and every final quantity (slope, ratio) is jackknifed at the end-of-pipeline level over leave-one-block-out samples; the annealed gate row's deviation from the exact operator is recorded per channel and added in quadrature as a systematic. Off-symmetry direction families are measured alongside the cubic stars, and the diamond-exclusion significance is computed in code for every channel, with the weakest channel quoted.

Gates. Every production run carries an annealed known-answer row through the identical pipeline; a failed gate discards the run. Amplitude gates exclude momenta near free-beat amplitude nodes for ratio estimators (shot-noise gating alone is insufficient there). Eigenvector work uses the $e^{+ik\cdot x}$ contraction (Sec. V). For interaction ratios, paired common-noise evolution (identical randomness with the imprint on and off) cancels the medium-ensemble variance, which is otherwise dominated by low-order conversion caustics.

Recorded failure modes. The following estimator pathologies were each encountered, diagnosed and firewalled during this program, and are listed because they generically afflict lattice pole measurements: (i) $\sqrt{\text{noise}}$ splitting at near-degenerate poles; (ii) exceptional-point misidentification in damped operators (real double eigenvalues accepted as propagating nodes); (iii) through-origin fits across a packet's death into the noise floor; (iv) ratio blowup at beat-amplitude nodes; (v) probing outside the (renormalized) linear cone window; (vi) branch mixing under scalar single-pole estimators at two-fold degenerate nodes; (vii) one-dimensional co-streaming substrates, whose companion locking invalidates absolute calibrations that do transfer in three dimensions.

Appendix D: Closed forms

Table II collects the closed forms. Two remarks. The in–out transfer's uniform modulus $1/D$ per read is a global scalar (every path reads one bit per sub-step) and is absorbed into wave-function renormalization; the physical statement is the vanishing of all *relative* damping. The quality factor of the in-state quasiparticle, $Q = \omega_0/\Gamma$, rises from 0.60 to 0.90 over $q \in [0.02, 0.25]$ and is bounded by κ per Theorem 3.

TABLE II. Exact expressions of the free and dressed spectra. All are verified against the operators they describe to at least 10^{-9}; $D = \sqrt{1-q} + \sqrt{q}$.

node modulus	$\lvert\lambda_0\rvert^2 = [(1-q)^2 + q^2]^6$
free width	$\Gamma(q) = -3\ln[(1-q)^2 + q^2]$
cone speed	$v(q)$: $2/\sqrt{3} \to 1.207$ (max at $q \simeq 0.30$)
dressing law (1D)	$v = 2\lvert 1-2q\rvert$
mass gap	$2m = 2\arctan[q_m/(1-q_m)]$
mass-layer modulus	$\sqrt{(1-q_m)^2 + q_m^2}$ per cycle
massive dispersion	$\cos(\omega - \omega_c) = \cos m \, \cos\varphi(k)$
interaction law	$U_g(k) = (1 - 2gq^2)\, U(k)$
Q-pinning constant	$\kappa = \pi/(2\ln 2) = 2.2662$
in–out transfer	$E_a(k)\,[\sqrt{1-q}\,\mathbb{1} + \sqrt{q}\,C_a]/D, \;\; \lvert\lambda\rvert \equiv 1/D$

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